1. Compute the length of $\mathbf{x} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$ using the dot product. Do the exercises using pen and paper.

   1 point

   - ○ 3
   - ◉ $\sqrt{11}$
   - ○ $\sqrt{3}$
   - ○ $\sqrt{5}$
   - ○ $\sqrt{13}$
   - ○ 11

2. Compute the angle (in rad) between $\mathbf{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ using the dot product.

   1 point

   2.99

3. Compute the distance between $\mathbf{x} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Do the exercises using pen and paper. Enter your answer as a decimal number (calculator is fine to get it).

   1 point

   5.38

4. Write a piece of code that computes the length of a given vector x.

   1 point

```python
import numpy as np

def length(x):
  """Compute the length of a vector"""
  length_x = np.sqrt(np.sum(x**2))

  return length_x

print(length(np.array([1,0])))
```

5. We are given two vectors

   1 point

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} -1 \\ 0 \\ 8 \end{bmatrix}$$

Compute the angle (in rad) between $\mathbf{x}$ and $\mathbf{x} - \mathbf{y}$.

Do the exercises using pen and paper, but you will need a calculator at some point.

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